Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

The change in the value of acceleration of earth towards sun, when the moon comes from the position of solar eclipse to the position on the other side of earth in line with sun is (Mass of moon $= 7.36 \times 10^{22}\ \mathrm{kg}$, the orbital radius of moon $3.8 \times 10^{8}\ \mathrm{m}$.

(A) $6.73 \times 10^{-2}\ \mathrm{m/s^2}$

(B) $6.73 \times 10^{-3}\ \mathrm{m/s^2}$

(C) $6.73 \times 10^{-4}\ \mathrm{m/s^2}$

(D) $6.73 \times 10^{-5}\ \mathrm{m/s^2}$

Q.2

A projectile is fired from the surface of earth of radius R with a velocity kv_e where v_e is the escape velocity and $k < 1$. Neglecting air resistance, the maximum height of rise will be

(A) $\frac{R}{k^2-1}$

(B) k^2R

(C) $\frac{k^2R}{1-k^2}$

(D) kR

Q.3

If the period of revolution of an artificial satellite just above the earth be T and the density of earth be ρ, then

(A) $\rho \mathrm{T}^2$ is a universal constant.

(B) $\rho \mathrm{T}^2$ varies with time

(C) $\rho \mathrm{T}^2 = \frac{3\pi}{\mathrm{G}}$

(D) Both (1) and (3)

Q.4

The mass and radius of earth and moon are $\mathrm{M}_1, \mathrm{R}_1$ and $\mathrm{M}_2, \mathrm{R}_2$ respectively. Their centres are d

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distance apart. With what velocity should a particle of mass m be projected from the mid point of their centres so that it may escape out to infinity -

(A) $\sqrt{\frac{\mathrm{G}(\mathrm{M}_1+\mathrm{M}_2)}{\mathrm{d}}}$

(B) $\sqrt{\frac{2\mathrm{G}(\mathrm{M}_1+\mathrm{M}_2)}{\mathrm{d}}}$

(C) $\sqrt{\frac{4\mathrm{G}(\mathrm{M}_1+\mathrm{M}_2)}{\mathrm{d}}}$

(D) $\sqrt{\frac{\mathrm{GM}_1\mathrm{M}_2}{\mathrm{d}}}$

Q.5

A satellite is revolving around the earth in a circular orbit of radius r and velocity V_0. A particle is projected from the satellite in forward direction with relative velocity $V=\left(\sqrt{\frac{7}{6}}-1\right)V_0$ (wrt state of satellite before projection), then the maximum distance from the earth's centre during the subsequent motion of the particle

(A) $\frac{5r}{3}$

(B) $\frac{7r}{5}$

(C) $\frac{7r}{4}$

(D) None

Q.6

If the satellite is stopped suddenly in its orbit which is at a distance = radius of earth from earth's surface and allowed to fall freely into the earth, the speed with which it hits the surface of earth will be -

(A) $7.919\ \mathrm{m/sec}$

(B) $7.919\ \mathrm{km/sec}$

(C) $11.2\ \mathrm{m/sec}$

(D) $11.2\ \mathrm{km/sec}$

Q.7

A projectile is fired vertically upward from the surface of earth with a velocity $\mathrm{Kv_e}$, where $\mathrm{v_e}$ is the escape velocity and $\mathrm{K}<1$. Neglecting air resistance, the maximum height to which it will rise measured from the centre of the earth is - (where are $R=$ radius of earth)

(A) $\frac{\mathrm{R}}{1-\mathrm{K}^2}$

(B) $\frac{\mathrm{R}}{\mathrm{K}^2}$

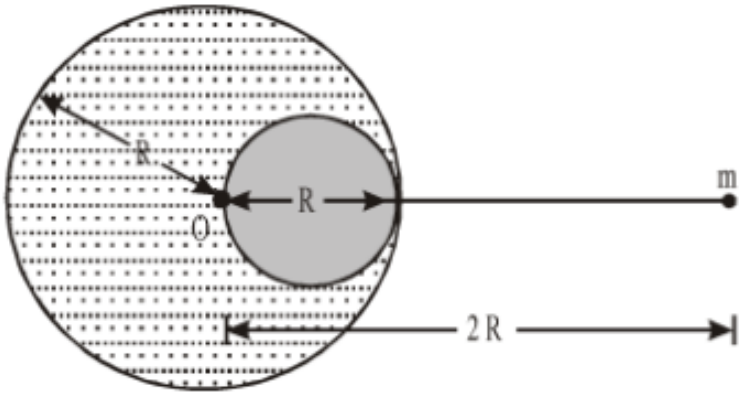
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(C) $\frac{1-K^2}{R}$

(D) $\frac{K^2}{R}$

Q.8

A uniform sphere of mass M and radius R exerts a force F on a small mass m situated at a distance of $2R$ from the centre O of the sphere. A spherical portion of diameter R is cut from the sphere as shown in figure. The force of attraction between the remaining part of the sphere and the mass m will be



(A) $\frac{7F}{9}$

(B) $\frac{2F}{3}$

(C) $\frac{4F}{9}$

(D) $\frac{F}{3}$

Q.9

A rocket starts vertically upwards with speed v_0. Choose the correct options (R is the radius of the earth)

(A) Speed v at a height is given by $v_0^2 - v^2 = \frac{2gh}{1+\frac{h}{R}}$

(B) The maximum height reached by the rocket fired with a speed of 90% of escape velocity is $2.26R$

(C) The maximum height reached by the rocket fired with a speed of 90% of escape velocity is 4.26R

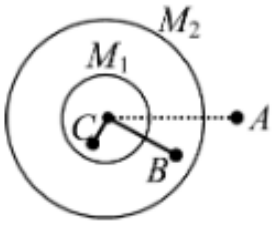
(D) Speed v at a height is given by $v_0^2 - v^2 = \frac{2gh}{1+\frac{h}{2R}}$

Q.10

Two concentric shells of uniform density of mass M_1 and M_2 are situated as shown in figure. The forces experienced by a particle of mass m when placed at position A, B and C respectively are

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$(OA = r_1, OB = r_2$ and $OC = r_3)$



(A) $F_A = \frac{G(M_1+M_2)m}{r_1^2}$

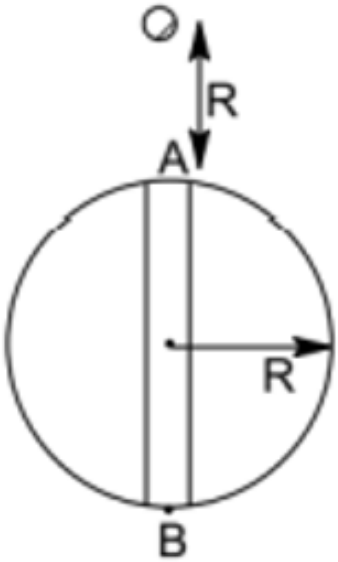
(B) $F_B = 0$

(C) $F_B = \frac{G(M_1)m}{r_2^2}$

(D) $F_C = 0$

Q.11

A small particle is dropped from a height R in front of a narrow tunnel dug inside the earth (along a diameter). Let M be the mass of earth, R be radius of earth. Let V_0, T be speed of particle when it reaches A and time taken by particle to go from A to B respectively. Assuming mass of particle to be negligible as compare to mass of earth, pick the correct option(s)



(A) $V_0 = \sqrt{\frac{GM}{2R}}$

(B) $V_0 = \sqrt{\frac{GM}{R}}$

(C) $\mathrm{T} = \frac{\pi}{2}\sqrt{\frac{\mathrm{R}^3}{\mathrm{GM}}}$

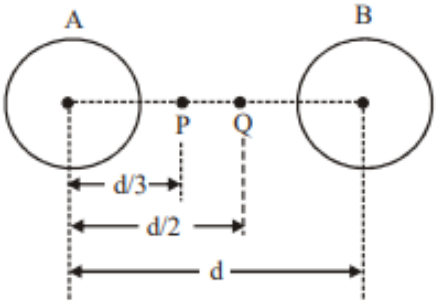
(D) $T = \pi\sqrt{\frac{R^3}{GM}}$

Q.12

Passage:

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A and B are two large spherical planets, d being distance between their centres. A body 1 placed at P and given a small push towards any planet (it does not matter which), eventually reaches that planet's sphere with velocity v_1. A body 2 placed at Q and given an initial velocity v_2 towards any planet (it does not matter which), eventually reaches one particular planet's surface with velocity v_3.



Question: Ratio of gravitational potentials at the surface A and B is $-$

(A) $= 1$

(B) > 1

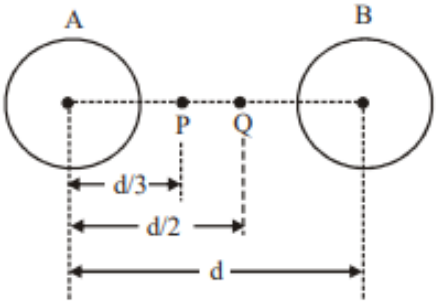
(C) < 1

(D) cannot be determined

Q.13

Passage:

A and B are two large spherical planets, d being distance between their centres. A body 1 placed at P and given a small push towards any planet (it does not matter which), eventually reaches that planet's sphere with velocity v_1. A body 2 placed at Q and given an initial velocity v_2 towards any planet (it does not matter which), eventually reaches one particular planet's surface with velocity v_3.



Question: Point of highest gravitational potential of the system is -

(A) Surface of A

(B) Surface of B

(C) P

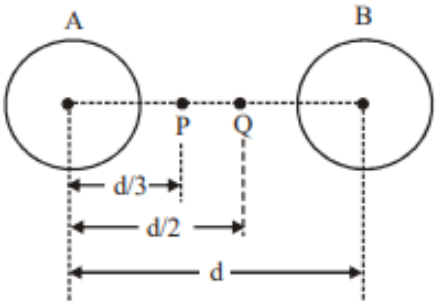
(D) Q

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Q.14

Passage:

A and B are two large spherical planets, d being distance between their centres. A body 1 placed at P and given a small push towards any planet (it does not matter which), eventually reaches that planet's sphere with velocity v_1. A body 2 placed at Q and given an initial velocity v_2 towards any planet (it does not matter which), eventually reaches one particular planet's surface with velocity v_3.



Question: Ratio of v_3 to v_1 is $-$

(A) $= 1$

(B) > 1

(C) < 1

(D) cannot be determined

Q.15

Passage:

If a tunnel is dug across the earth passing through centre and a particle is drop from surface it perform SHM inside tunnel with time period, $\mathrm{T} = 2\pi\sqrt{\frac{\text{Radius}}{\mathrm{g}}}$

Question: Assume that a tunnel is dug across the earth (radius $= R$) passing through its centre. Find the time, a particle takes to reach centre of the earth, if it is projected into the tunnel from surface of earth with speed needed for it to escape the gravitational field of earth.

(A) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\sqrt{\frac{\mathrm{R_e}}{\mathrm{g}}}$

(B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\sqrt{\frac{2\mathrm{R_e}}{\mathrm{g}}}$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\sqrt{\frac{\mathrm{R_e}}{2\,\mathrm{g}}}$

(D) None of these

Q.16

Passage:

If a tunnel is dug across the earth passing through centre and a particle is drop from surface it perform

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SHM inside tunnel with time period, $T = 2\pi\sqrt{\frac{\text{Radius}}{g}}$

Question: If a particle is drop from height $h(h << R)$ into the tunnel then choose the correct option $-$

(A) Particle will perform SHM

(B) Particle will not perform SHM

(C) Particle will perform periodic motion

(D) Both (2) and (3)

Q.17

Passage:

If a tunnel is dug across the earth passing through centre and a particle is drop from surface it perform SHM inside tunnel with time period, $T = 2\pi\sqrt{\frac{\text{Radius}}{g}}$

Question: Find the time period for the motion in previous question

(A) $2\pi\sqrt{\frac{R}{g}} + 4\sqrt{\frac{2h}{g}}$

(B) $2\pi\sqrt{\frac{R}{g}} + 2\sqrt{\frac{2h}{g}}$

(C) $\pi\sqrt{\frac{R}{g}} + 2\sqrt{\frac{2h}{g}}$

(D) None of these

Q.18

We take three configurations 1 hollow sphere, 1 solid sphere and 1 solid sphere inside a hollow sphere (both concentric with different radii). Each sphere is of mass m, V and E are the gravitational potential and field at a distance r from the centre. Then-

Column I	**Column II**
(A) The plot of V versus r for solid sphere.	(p) discontinuous
(B) The plot of E versus r for spherical shell.	(q) continuous
(C) The plot of V versus r for solid sphere inside a hollow sphere.	(r) differentiable
(D) The plot of E versus r for solid sphere.	(s) non-differentiable

(A) A-(qs) B-(q) C-(sqr) D-(rq)

(B) A-(qs) B-(r) C-(qs) D-(qr)

(C) A-(qsr) B-(q) C-(sq) D-(p)

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(D) A-(pqr) B-(qs) C-(qs) D-(p)

Q.19

Submarines on Europa. Some scientists are eager to send a remote-controlled submarine to Jupiter's moon Europa to search for life in its oceans below an icy crust. Europa's mass has been measured to be $4.80 \times 10^{22}\ \text{kg}$, its diameter is $3120\ \text{km}$, and it has no appreciable atmosphere. Assume that the layer of ice at the surface is not thick enough to exert substantial force on the water. If the windows of the submarine you are designing each have an area of $625\ \text{cm}^2$ and can stand a maximum inward force of $8750\ \text{N}$ per window, what is the greatest depth to which this submarine can safely dive?

Q.20

If an object having mass $\sqrt{3}\ \text{kg}$ is moving with a velocity that is $\frac{1}{2}$ the velocity of light then inertial mass (in kg) will be -

Answers & Solutions

Q.1 Answer:

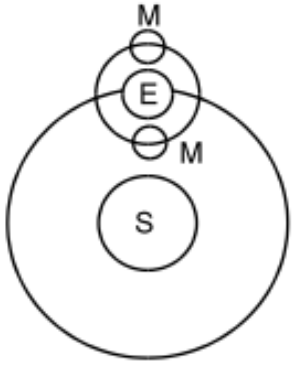
$6.73 \times 10^{-5}\ \text{m/s}^2$

Solution:

In the position of solar eclipse, net force on earth $\text{F}_\text{E} = \text{F}_\text{M} + \text{F}_\text{S}$ **In the position of lunar eclipse, net force on earth** $\text{F}'_\text{E} = \text{F}_\text{S} - \text{F}_\text{M}$

∴ Change in acceleration of earth,

$\Delta \text{f} = \frac{2\text{GM}}{\text{R}^2} = \frac{2 \times 6.67 \times 10^{-11} \times 7.36 \times 10^{22}}{3.82^2 \times 10^{16}} = 6.73 \times 10^{-5}\ \text{m/s}^2$



Q.2 Answer:

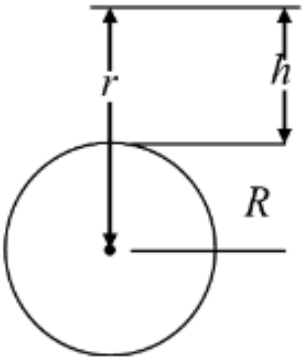
$\frac{k^2 R}{1-k^2}$

Solution:

By conservation of energy,

$\frac{1}{2}mv^2 = \frac{mgh}{\left(1+\frac{h}{R}\right)}$

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here $v = kv_e = k\sqrt{2gR}$

$\therefore \quad \frac{1}{2}m(k\sqrt{2gR})^2 = \frac{mgh}{1+h/R}$

$h = \frac{k^2R}{1-k^2}$

Q.3 Answer:

Both (1) and (3)

Solution:

If the period of revolution of a satellite about the earth be T, then

$\mathrm{T}^2 = \frac{4\pi^2(\mathrm{R_e}+\mathrm{h})^3}{\mathrm{GM_e}}$, **where** h **is the height of the satellite from earth's surface.**

$\therefore \quad M_e = \frac{4\pi^2(R_e+h)^3}{GT^2}$

The satellite is revolving just above the earth, hence h **is negligible compared to** $\mathrm{R_e}$.

$\therefore \quad \mathrm{M_e} = \frac{4\pi^2\mathrm{R_e}^3}{\mathrm{GT}^2}$

But $\mathrm{M_e} = \frac{4}{3}\pi\mathrm{R_e^3}\rho$ **where** ρ **is the density of the earth. Thus**

$\frac{4}{3}\pi\mathrm{R_e^3}\rho = \frac{4\pi^2\mathrm{R_e}^3}{\mathrm{GT}^2} \quad \therefore \quad \rho\mathrm{T}^2 = \frac{3\pi}{\mathrm{G}}$

which is universal constant.

To determine its value, $\rho\mathrm{T}^2 = \frac{3\pi}{\mathrm{G}} = \frac{3\times3.14}{6.67\times10^{-11}\ \mathrm{m^3/kg-s^2}}$

Q.4 Answer:

$\sqrt{\frac{4\mathrm{G}(\mathrm{M_1}+\mathrm{M_2})}{\mathrm{d}}}$

Solution:

On conserving mechanical energy

Initial total energy = final total energy

$$\frac{-\mathrm{GM_2\ m}}{\mathrm{d}/2} - \frac{\mathrm{GM_1\ m}}{\mathrm{d}/2} = \frac{1}{2}\mathrm{mV}^2$$

$$\Rightarrow \quad \mathrm{V}^2 = \frac{4\mathrm{G}}{\mathrm{d}}(\mathrm{M_1}+\mathrm{M_2})$$

$$\Rightarrow \quad \mathrm{V} = \sqrt{\frac{4\mathrm{G}(\mathrm{M_1}+\mathrm{M_2})}{\mathrm{d}}}$$

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Q.5 Answer:

$\frac{7r}{5}$

Solution:

$$V_{\text{absolute}} = V_{RV} + V_0$$
$$= \left(\sqrt{\frac{7}{6}} - 1\right) \mathrm{V_0} + \mathrm{V_0} = \sqrt{\frac{7}{6}}\,\mathrm{V_0}$$

Outer ellipse

Apply conservation of angular momenta $\mathrm{Li = Lf,\ \sqrt{\frac{7}{6}} mvv_0 r = mvr_{mm}}$

$v_0 = \sqrt{\frac{GM}{r}}$

Apply conservation of Total energy

$\frac{1}{2}mV_0^2 - \frac{GMm}{R} = \frac{1}{2}mV^2 - \frac{GMm}{r_{\max}}$

Solve the quadratic equation

Then $r_{\max} = 7\mathrm{r}/5$

Q.6 Answer:

$7.919\ \mathrm{km/sec}$

Solution:

At $\mathrm{h = R}$**, Kinetic Energy is zero as satellite is stopped suddenly.**

$\mathrm{T \cdot E_h = 0 + \left(-\frac{GMm}{2R}\right) \quad (r = h + R = 2R)}$

At the surface of earth, let velocity attained be v**.**

$\mathrm{T \cdot E_s = \frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right)}$

Apply conservation of energy at h **and surface of earth-**

$\mathrm{-\frac{GMm}{2R} = \frac{1}{2}mv^2 - \frac{GMm}{R}}$

$\mathrm{\Rightarrow v = \sqrt{\frac{GM}{R}}}$

$\mathrm{v = \sqrt{gR}}$

$\mathrm{v = \sqrt{9.8 \times 6400 \times 1000}}$

$\mathrm{\Longrightarrow v = 7.919\ km/s}$

Q.7 Answer:

$\frac{\mathrm{R}}{1-\mathrm{K}^2}$

Solution:

If v_e **is the escape velocity then**

$\mathrm{\frac{1}{2}v_e^2 = \frac{GMm}{R}}$

Now the projectile is fired with speed $\mathrm{kv_e}$**.**

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Then loss in kinetic energy is gain in potential energy

$$\Rightarrow \frac{1}{2}\,\mathrm{m}\left(\mathrm{kv_e}\right)^2 = \mathrm{GMm}\left(\frac{1}{\mathrm{R}} - \frac{1}{\mathrm{h}}\right)$$

$$\Rightarrow \mathrm{k}^2\left(\frac{\mathrm{GMm}}{\mathrm{R}}\right) = \mathrm{GMm}\left(\frac{1}{\mathrm{R}} - \frac{1}{\mathrm{h}}\right)$$

$$\Rightarrow \mathrm{h} = \frac{\mathrm{R}}{1-\mathrm{k}^2}$$

Q.8 Answer:

$\frac{7F}{9}$

Solution:

$F = \frac{GmM}{(2R)^2} = \frac{GmM}{4R^2}$ **...(i)**

Mass of complete sphere is $M = \frac{4\pi}{3}R^3\rho$.

Mass of the cut out portion is $\mathrm{m_0} = \frac{4\pi}{3}\left(\frac{R}{2}\right)^2\rho$.

Thus, $m_0 = \frac{M}{8}$.

The distance between the centre of the cut out portion and mass $m = 2R - \frac{R}{2} = \frac{3R}{2}$.

Hence the force of attraction between the cut out portion and mass m **is**

$f = \frac{Gm_0m}{(3R/2)^2} = \frac{G(M/8)m}{9R^2/4} = \frac{GmM}{4R^2} \times \frac{2}{9}$

Using (i), we get $f = \frac{2F}{9}$.

Therefore, the force of attraction between the remaining part of the sphere and mass $m =$

$F - f$

$= F - \frac{2F}{9} = \frac{7F}{9}$

Q.9 Answer:

Speed v at a height is given by $v_0^2 - v^2 = \frac{2gh}{1+\frac{h}{R}}$

Solution:

Kinetic energy on the surface of earth $= \frac{1}{2}\mathrm{mv_0^2}$

Potential energy on the surface of earth $= \frac{-\mathrm{GMm}}{\mathrm{R}}$

Total energy $= \frac{1}{2}\mathrm{mv_0^2} - \frac{\mathrm{GMm}}{\mathrm{R}}$;

Kinetic energy at a height $\mathrm{h} = \frac{1}{2}\mathrm{mv^2}$

Potential energy at this height $= \frac{-\mathrm{GMm}}{(\mathrm{R+h})}$;

Total energy $= \frac{1}{2}\mathrm{mv^2} - \frac{\mathrm{GMm}}{\mathrm{R+h}}$

By the principle of conservation of energy, $\frac{1}{2}\mathrm{mv^2} - \frac{\mathrm{GMm}}{\mathrm{R+h}} = \frac{1}{2}\mathrm{mv_0^2} - \frac{\mathrm{GMm}}{\mathrm{R}}$; $\frac{1}{2}\left(\mathrm{v_0^2} - \mathrm{v^2}\right) = \frac{\mathrm{GM}}{\mathrm{R}} - \frac{\mathrm{GM}}{\mathrm{R+h}}$.

But $\mathrm{GM} = \mathrm{gR^2}$

$\therefore \quad \frac{1}{2}\left(v_0^2 - v^2\right) = \frac{gR^2h}{R(R+h)}; v_0^2 - v^2 = \frac{2gRh}{R+h} = \frac{2gh}{1+\frac{h}{R}}$

At maximum height $\mathrm{v} = 0$.

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The initial velocity $v_0 = (90\%)v_{escape} = 0.9\sqrt{2gR}$ **then** $(0.9\sqrt{2gR})^2 - 0 = \frac{2gh_{max}}{1+\frac{h_{max}}{R}}; \quad 0.81R = \frac{h_{max}}{1+\frac{h_{max}}{R}}$

$0.81R + 0.81\ h_{max} = h_{max}; 0.19\ h_{max} = 0.81R$ **Maximum height reached by the rocket**

$h_{max} = \frac{0.81R}{0.19} = 4.26R$

Q.10 Answer:

$F_A = \frac{G(M_1+M_2)m}{r_1^2}$

Solution:

$$F_A = \frac{C_2\left(M_1+M_2\right)m}{r_1^2}, F_B = \frac{GM_1m}{r_2^2}, F_C = 0$$

$\therefore \quad (1)\ (3)$ and (4) is the correct answer

Q.11 Answer:

$V_0 = \sqrt{\frac{GM}{R}}$

Solution:

$$\frac{-GMm}{2R} = \frac{-GMm}{R} + \frac{1}{2}mv_0^2$$

$$\Rightarrow v_0 = \sqrt{\frac{GM}{R}}$$

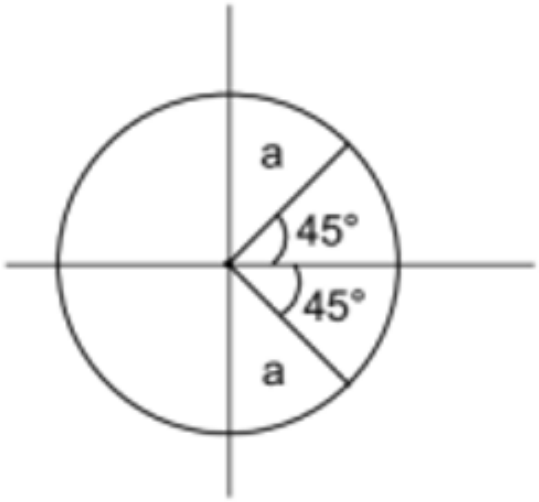
While in the tunnel;

$$\frac{d^2x}{dt^2} = -\left(\frac{GM}{R^3}\right)x$$

$$v_0^2 = \frac{GM}{R} = \omega^2\left(a^2 - R^2\right) \Rightarrow a = \sqrt{2}R$$

$$\Rightarrow T_{A\to B} = \frac{1}{4} \times 2\pi\sqrt{\frac{R^3}{GM}}$$

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Q.12 Answer:

$= 1$

Solution:

Clearly body at P **is in equilibrium**

$\frac{\mathrm{Gm_A m}}{(\mathrm{d}/3)^2} = \frac{\mathrm{Gm_B m}}{(2\,\mathrm{d}/3)^2} \Rightarrow \mathrm{m_B} = 4\,\mathrm{m_A}$

Let U_A, U_P, U_Q **and** U_B **refers to potential energies at surface** A**, at** P**, at** Q **and at surface of** B **respectively.**

Then data 1: $\mathrm{U_P} = \mathrm{U_A} + \frac{1}{2}\mathrm{mv_1^2}$ **and** $\mathrm{U_P} = \mathrm{U_B} + \frac{1}{2}\mathrm{mv_1^2} \ldots (1)$

$\Rightarrow \quad U_A = U_B$ **and clearly** U_P **is maximum.**

Q.13 Answer:

P

Solution:

Clearly body at P **is in equilibrium**

$\frac{\mathrm{Gm_A m}}{(\mathrm{d}/3)^2} = \frac{\mathrm{Gm_B m}}{(2\,\mathrm{d}/3)^2} \Rightarrow \mathrm{m_B} = 4\,\mathrm{m_A}$

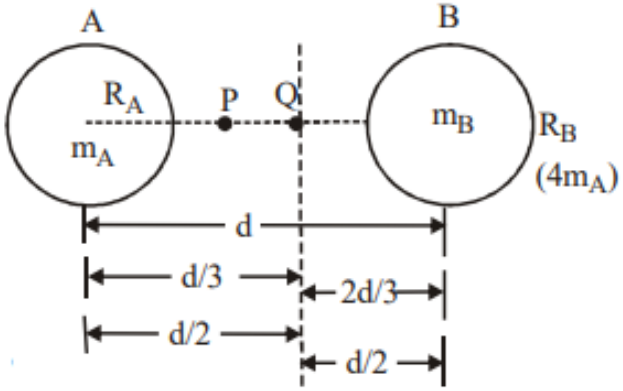
Let U_A, U_P, U_Q **and** U_B **refers to potential energies at surface** A**, at** P**, at** Q **and at surface of** B **respectively.**

Then data 1: $\mathrm{U_P} = \mathrm{U_A} + \frac{1}{2}\mathrm{mv_1^2}$ **and** $\mathrm{U_P} = \mathrm{U_B} + \frac{1}{2}\mathrm{mv_1^2} \ldots (1)$

$\Rightarrow \quad U_P = U_B$ **and clearly** U_P **is maximum.**

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$\Rightarrow$ p is point of highest gravitational potential.



Q.14 Answer:

< 1

Solution:

If body 2 is given v_2 towards A, and if $\frac{1}{2}mv_2^2 > U_P - U_Q$, then it will cross P with a non-zero velocity and will reach A.

If $\frac{1}{2}mv_2^2 < U_P - U_Q$, it will reach B.

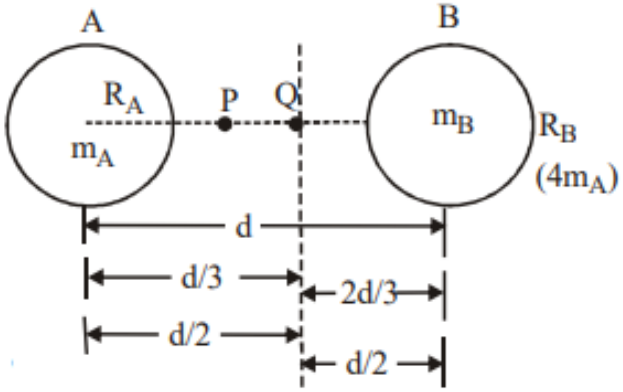
Hence the particular planet is B, and $\frac{1}{2}mv_2^2 < U_P - U_Q$

$\therefore \frac{1}{2}mv_2^2 + U_Q < U_P = U_B + \frac{1}{2}mv_1^2$

[Using (1)]

But $\frac{1}{2}mv_2^2 + U_Q = \frac{1}{2}mv_3^2 + U_B$ (By conservation of energy)

$\Rightarrow U_B + \frac{1}{2}mv_3^2 < U_B + \frac{1}{2}mv_1^2 \Rightarrow v_3 < v_1 \Rightarrow \frac{v_3}{v_1} < 1$



Q.15 Answer:

$\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\sqrt{\frac{R_e}{g}}$

Solution:

We know if a particle is released (initial speed zero) from earth surface time period $= 2\pi\sqrt{\frac{\text{Radius}}{g}}$, so we imagine a earth of radius A from where particle is released and when it reaches at distance R from earth centre its speed is v_e.

Mass of assumed earth of radius is A.

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$M' = \left(\frac{A}{R}\right)^3 M$

According to energy conservation

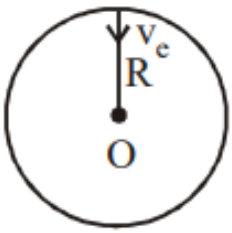
$-\frac{GM'm}{A} + 0 = -\frac{GM'}{2A^3}\left(3A^2 - R^2\right)m + \frac{1}{2}m\left(\frac{2GM}{R}\right)$

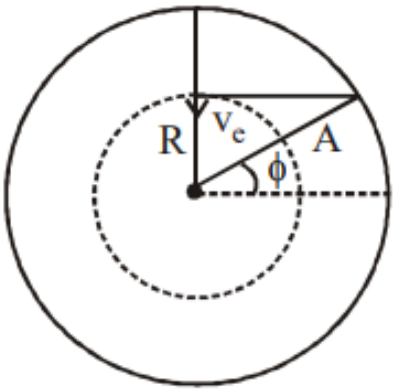
$-\frac{A^3}{R^3}\frac{M}{A} = -\frac{A^3}{R^3}\frac{M}{2A^3}\left(3A^2 - R^2\right) + \frac{M}{R}$

$-A^2 = -\frac{3}{2}A^2 + \frac{R^2}{2} + R^2; A = \sqrt{3}R$

$\sin\phi = \frac{1}{\sqrt{3}}; \phi = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$T = \left(\frac{\phi}{2\pi}\right) T = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\sqrt{\frac{A}{g_A}} = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\sqrt{\frac{R_e}{g}}$





Q.16 Answer:

Both (2) and (3)

Solution:

At a height h from earth surface acceleration due to gravity is not proportional to distance from centre hence particle will not perform SHM but the motion will be periodic.

Q.17 Answer:

$2\pi\sqrt{\frac{R}{g}} + 4\sqrt{\frac{2h}{g}}$

Solution:

$h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}}$

Time period $= 2\pi\sqrt{\frac{R}{g}} + 4t = 2\pi\sqrt{\frac{R}{g}} + 4\sqrt{\frac{2h}{g}}$

Q.18 Answer:

A-(qs) B-(r) C-(qs) D-(qr)

Solution:

$(A) - \mathbf{qs}, (B) - \mathbf{r}, (C) - \mathbf{qs}, (D) - \mathbf{qr}.$

(A) The plot of V versus r for solid sphere is both continuous and differentiable

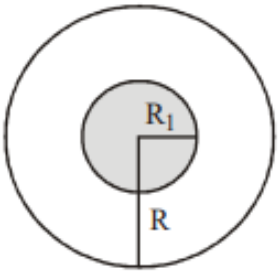
(B) The plot of E versus r for spherical shell is discontinuous and so non-differentiable

(C) The plot of V versus r for solid sphere inside a hollow sphere is continuous but

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nondifferentiable

(D) The plot of E versus r for solid sphere is continuous but differentiable



Q.19 Answer:

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Solution:

Identify: The information about Europa allows us to evaluate g at the surface of Europa. Since there is no atmosphere, $p_0 = 0$ at the surface. The pressure at depth h is $p = \rho g h$. The imward force on the window is $F_\perp = pA$.

SET UP: $g = \frac{Gm}{R^2}$, where $G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$ $R = 1.560 \times 10^6\ \mathrm{m}$. Assume the ocean water has density $\rho = 1.00 \times 10^3\ \mathrm{kg/m^3}$.

EXECUTE: $g = \frac{\left(6.67\times10^{-11}\ \mathrm{N\cdot m^2/kg^2}\right)\left(4.80\times10^{22}\ \mathrm{kg}\right)}{(1.560\times10^6\ \mathrm{m})^2} = 1.3156\ \mathrm{m/s^2}$. The maximum pressure at the window is $p = \frac{8750\ \mathrm{N}}{(0.250\ \mathrm{m})^2} = 1.40 \times 10^5\ \mathrm{Pa}$. $p = \rho g h$ so for h we get $h = \frac{1.40\times10^5\ \mathrm{Pa}}{(1.00\times10^3\ \mathrm{kg/m^3})(1.3156\ \mathrm{m/s^2})} = 106\ \mathrm{m}$

Q.20 Answer:

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Solution:

$\mathrm{m_g} = \sqrt{3}\ \mathrm{kg}, \mathrm{v} = \frac{\mathrm{c}}{2}$

$\mathrm{m} = \frac{\mathrm{m_0}}{\sqrt{1-(\mathrm{v^2/c^2})}} = \frac{\sqrt{3}}{\sqrt{1-\frac{\mathrm{c^2}}{4\mathrm{xc^2}}}} = \frac{\sqrt{3}}{\frac{\sqrt{3}}{2}}\ \mathrm{kg} = 2\ \mathrm{kg}$

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